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Cryptocurrency Perpetual Futures and Swaps: A Systematic Literature Review

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Abstract

This systematic literature review synthesizes research on cryptocurrency perpetual futures and perpetual swaps. It examines 22 substantive studies, comprising 15 peer-reviewed publications and 7 academic working papers, with emphasis on work published from 2021 to April 2026, while retaining directly relevant foundational contributions. Reported in line with PRISMA 2020, the review asks how the literature defines perpetual contracts, explains pricing and funding mechanisms, characterizes basis behavior and convergence, evaluates market quality and price discovery, analyzes leverage and liquidation, compares centralized and decentralized exchange designs, and addresses legal classification and regulation. The evidence indicates that pricing and no-arbitrage theory are the most mature strands of research. Peer-reviewed studies generally find that funding-based design reduces rollover frictions and improves spot tracking relative to dated futures. Yet perpetuals do not eliminate risk. They relocate risk into funding windows, leverage choice, liquidation cascades, informed trading, venue architecture, and regulatory fragmentation. The empirical literature is strongest on market microstructure and risk management, but remains thinner on trader behavior and systematic CEX–DEX comparison. Legal scholarship shows persistent instability in classification and cross-border arbitrage. The review concludes that the next research frontier lies in funding-rule design, spot-market spillovers, stress transmission, and exchange-architecture comparison.

Keywords: perpetual futures, perpetual swaps, cryptocurrency derivatives, funding rate, basis, market quality, liquidation, exchange design, regulation

JEL classification: G13, G14, G15, G18

Introduction

This systematic literature review is about perpetuals. The motivation for this study is that perpetuals dominate crypto derivatives markets in both scale and usage (Amina Bank, 2025; CoinGecko, 2026; CoinGlass, 2025; CME Group, 2026; Shi et al., 2024). In Q3 2025, perpetuals accounted for about 78% of total crypto derivatives trading activity, which indicates a strong market preference for non-expiring contracts (Amina Bank, 2025). CoinGecko (2026) reports a 2025 volume of \$92.9 trillion across leading perpetual venues, including \$6.7 trillion on perpetual DEXs. CME Group (2026) reports that average daily volume in its listed cryptocurrency futures and options rose 139% to 278,000 contracts in 2025. That figure does not measure perpetuals directly, but it does indicate broader growth in crypto derivatives activity (Amina Bank, 2025; CoinGecko, 2026; CoinGlass, 2025; CME Group, 2026). CoinGlass (2025) estimates total crypto derivatives volume at \$85.7 trillion across instruments. Although these datasets use different coverage, they indicate that perpetuals account for a high share of market turnover (Amina Bank, 2025; CoinGecko, 2026; CoinGlass, 2025; CME Group, 2026).

Neubert et al. (2026) define perpetuals as follows:

Perpetuals are derivative contracts on an underlying asset that (a) have no contractual maturity, (b) are marked to market through an exchange-defined mark price, (c) require margin collateral and allow leverage, and (d) include periodic funding payments between longs and shorts that depend on the perpetual–spot spread and often include an interest-rate component. (Neubert et al., 2026, pp. 3–4)

The purpose of this study is to follow Neubert et al.'s (2026) call for further research to shed more light on the state of research on cryptocurrency perpetual futures and swaps.

This review article asks one question: What is the current state of research on cryptocurrency perpetual futures and perpetual swaps, and how should that research be organized conceptually, methodologically, and empirically? To answer that question, the article addresses seven linked issues. It reviews definitions and taxonomy. It synthesizes pricing, funding, no-arbitrage, basis behavior, and convergence. It evaluates evidence on liquidity, market quality, price discovery, leverage, liquidation, hedging, trader behavior, exchange design, and regulation.

The review makes three contributions. First, it offers an integrated taxonomy of perpetual contracts and related instruments. Second, it distinguishes results that now rest on peer-reviewed evidence from claims that still depend on working papers. Third, it develops a unifying argument: perpetuals appear to have become dominant in part because the funding mechanism reduces rollover frictions and generally compresses basis risk, but the same design redistributes risk into new channels, especially liquidation, informed trading, venue architecture, and regulatory arbitrage (De Blasis & Webb, 2022; Gornall et al., 2025; Cheng et al., 2021; Ruan & Streltsov, 2022; Phang, 2023; Dell’Erba & Vianelli, 2025).

The article proceeds as follows. Section 2 explains the review method and evidence base. Section 3 clarifies the conceptual foundations and taxonomy. Sections 4 to 7 synthesize the finance and market-design literatures. Section 8 reviews the legal and regulatory debate. Section 9 integrates the evidence and develops a research agenda. Section 10 concludes.

Method

Search strategy and eligibility criteria

This review was reported in accordance with PRISMA 2020, a reporting guideline for systematic reviews (Page et al., 2021). The operational design also drew on recent guidance for systematic literature reviews in applied business and management research (Neubert, 2022a,

2022b, 2026). The search covered publications from January 2021 to April 2026 and was updated in April 2026. The review searched SSRN, RePEc, ScienceDirect, Wiley Online Library, SpringerLink, Taylor & Francis Online, JSTOR, Google Scholar, and publisher pages identified through citation chaining. The core search string was: ("perpetual future*" OR "perpetual swap*" OR "perpetual contract*" OR "inverse perpetual*" OR "linear perpetual*" OR "quanto perpetual*" OR "power perpetual*" OR "everlasting option*") AND (cryptocurrency OR crypto OR bitcoin OR BTC OR ethereum OR ETH OR stablecoin OR DeFi OR "decentralized exchange*" OR CEX OR DEX) AND (pricing OR "funding rate" OR basis OR arbitrage OR "market quality" OR liquidity OR microstructure OR "price discovery" OR hedging OR leverage OR liquidation OR margin OR regulation OR law OR "investor behavior" OR trader*). The syntax was adapted to each source's search interface and supplemented with backward- and forward-citation chaining.

The review was not registered in a public registry, and no standalone protocol was published. However, the eligibility criteria, extraction fields, and thematic coding scheme were specified prior to full-text screening. They were applied consistently throughout the selection and synthesis.

Studies were included if they focused directly on cryptocurrency perpetual futures or perpetual swaps, or if they devoted a substantial section to perpetuals within a broader crypto-derivatives analysis. Eligible studies had to make an original empirical, theoretical, legal, or comparative contribution. Peer-reviewed journal articles were prioritized for the 2021-2026 period, while academic working papers were retained when they addressed frontier issues not yet covered in the journal literature. Title/abstract screening and full-text screening followed the

same inclusion and exclusion rules and were resolved by author discussion when the scope was ambiguous.

PRISMA flow and screening

Table 1 reports the PRISMA flow. The identification stage yielded 52 records from the structured database, repository, publisher, and citation searches. Deduplication removed 11 records, leaving 41 records for title and abstract screening. The screening stage excluded 10 records that did not meet the topical inclusion criteria. Full-text assessment was conducted on the remaining 31 records. Nine full-text records were excluded: four lacked substantial perpetual-contract analysis, two were methodological or background papers rather than substantive studies, two duplicated a newer or more complete version of the same study, and one was non-academic commentary.

The final substantive evidence base contains 22 studies (see Table 3). Fifteen are peer-reviewed publications. Seven are academic working papers. The peer-reviewed subset supplies the strongest evidence on pricing, hedging, liquidation, market microstructure, and legal comparison (Ackerer et al., 2025; Alexander et al., 2023; Cheng et al., 2021; Phang, 2023). The working-paper subset remains important because several frontier themes, especially basis-risk evidence, market-quality effects, and regulatory fragmentation, are still represented primarily by recent working papers (Christin et al., 2023; Dell’Erba & Vianelli, 2025; Gornall et al., 2025; Ruan & Streltsov, 2022). Additional methodological references, such as Page et al. (2021) and Neubert (2022a, 2022b, 2026), support the review design but are not counted in the substantive evidence base.

Data extraction and synthesis

For each included study, the review extracted the following fields: APA citation, year, outlet type, research design, data source and sample period, exchange, asset, contract type, main research question, method or model, key findings, limitations, and thematic classification. Each study was assigned one primary theme, while secondary themes and methodological or data-quality issues were recorded qualitatively. The primary themes were: foundations and definitions; pricing, funding, and no-arbitrage; basis, convergence, and arbitrage; market quality, liquidity, and microstructure; price discovery and spot-futures interaction; hedging, leverage, liquidation, and margin; investor behavior and speculative demand; CEX versus DEX market design; and regulation, law, and policy.

Study appraisal was qualitative rather than score-based (Neubert, 2025). Each included study was assessed on four dimensions: publication status, methodological transparency, data provenance, and direct relevance to perpetual contracts. No single numeric risk-of-bias tool was imposed because the evidence base combines analytical finance, empirical microstructure, optimization, and legal-comparative research. Instead, the synthesis weights peer-reviewed evidence more heavily and reports working-paper findings more cautiously where replication or publication is still pending.

Methodological heterogeneity is substantial (Ackerer et al., 2025; Alexander et al., 2024; Cheng et al., 2021; Phang, 2023; Dell’Erba & Vianelli, 2025). Pricing and no-arbitrage studies are mainly analytical and derive closed-form or random-time valuation results (Ackerer et al., 2025; Alexander et al., 2024; Cheng et al., 2021; Phang, 2023; Dell’Erba & Vianelli, 2025). Market-quality and price-discovery studies are primarily quantitative and rely on high-frequency spot, futures, funding-rate, or order-book data (Ackerer et al., 2025; Alexander et al., 2024;

Cheng et al., 2021; Phang, 2023; Dell’Erba & Vianelli, 2025). Risk-management studies are mixed-method because they combine modeling, calibration, and exchange-level evidence (Ackerer et al., 2025; Alexander et al., 2024; Cheng et al., 2021; Phang, 2023; Dell’Erba & Vianelli, 2025). The law-and-policy literature is qualitative, doctrinal, and comparative (Ackerer et al., 2025; Alexander et al., 2024; Cheng et al., 2021; Phang, 2023; Dell’Erba & Vianelli, 2025). This heterogeneity is substantively useful because it covers multiple dimensions of the instrument, but it limits direct study-to-study comparability and makes a quantitative meta-analysis inappropriate (Ackerer et al., 2025; Alexander et al., 2024; Cheng et al., 2021; Phang, 2023; Dell’Erba & Vianelli, 2025). Table 2 reports the primary-theme allocation only; methodological features and data-quality issues are treated as cross-cutting secondary attributes rather than as a separate primary category.

Table 1*PRISMA 2020 flow summary*

Stage	Count
Records identified through structured database, repository, publisher, and citation searches	52
Duplicate records removed	11
Title/abstract records screened	41
Records excluded at title/abstract screening	10
Full-text records assessed for eligibility	31
Full-text exclusions: no substantial perpetual-contract analysis	4
Full-text exclusions: methodological/background only	2
Full-text exclusions: duplicate or superseded version	2
Full-text exclusions: non-academic commentary	1
Studies included in the review	22

Table 2*Evidence profile by theme and evidence type*

Theme	Peer-reviewed	Working paper	Total
Basis and arbitrage	3	1	4
CEX vs. DEX design	1	0	1
Foundations	1	1	2
Hedging and liquidation	4	0	4
Investor behavior	0	1	1
Market quality	1	1	2
Price discovery	2	0	2
Pricing and no-arbitrage	2	2	4
Regulation and policy	1	1	2
Total	15	7	22

Table 3*Reviewed studies included in the systematic literature review*

No.	Study	Evidence type	Outlet	Method
1	Ackerer et al. (2025). <i>Perpetual futures pricing.</i>	Peer-reviewed	<i>Mathematical Finance</i>	Modeling
2	Alexander et al. (2024). <i>Arbitrage opportunities and efficiency tests in crypto derivatives.</i>	Peer-reviewed	<i>Journal of Financial Markets</i>	Empirical
3	Alexander et al. (2020). <i>BitMEX bitcoin derivatives: Price discovery, informational efficiency, and hedging effectiveness.</i>	Peer-reviewed	<i>Journal of Futures Markets</i>	Empirical
4	Alexander et al. (2023). <i>Hedging with automatic liquidation and leverage selection on Bitcoin futures.</i>	Peer-reviewed	<i>European Journal of Operational Research</i>	Mixed
5	Angeris et al. (2023). <i>Short communication: A primer on perpetuals.</i>	Peer-reviewed	<i>SIAM Journal on Financial Mathematics</i>	Modeling
6	Atzberger et al. (2024). <i>A low-volatility strategy based on hedging a quanto perpetual swap on BitMEX.</i>	Peer-reviewed	<i>IEEE ICBC</i>	Empirical
7	Cheng et al. (2021). <i>Liquidation, leverage, and optimal margin in bitcoin futures markets.</i>	Peer-reviewed	<i>Applied Economics</i>	Mixed
8	Chen et al. (2024). <i>Perpetual future contracts in centralized and decentralized exchanges: Mechanism and traders' behavior.</i>	Peer-reviewed	<i>Electronic Markets</i>	Mixed
9	Clark (2023). <i>Spanning with power perpetuals.</i>	Working paper	<i>SSRN</i>	Modeling
10	Christin et al. (2023). <i>The crypto carry trade.</i>	Working paper	<i>Carnegie Mellon University working paper</i>	Empirical

11	De Blasis and Webb (2022). <i>Arbitrage, contract design, and market structure in Bitcoin futures markets.</i>	Peer-reviewed	<i>Journal of Futures Markets</i>	Empirical
12	Dell'Erba and Vianelli (2025). <i>Crypto-derivatives regulation(s).</i>	Working paper	<i>SSRN</i>	Legal
13	Deng et al. (2021). <i>Optimal bitcoin trading with inverse futures.</i>	Peer-reviewed	<i>Annals of Operations Research</i>	Mixed
14	Giagkiozis and Said (2024). <i>Reconciling open interest with traded volume in perpetual swaps.</i>	Peer-reviewed	<i>Ledger</i>	Empirical
15	Gornall et al. (2025). <i>Perpetual futures and basis risk: Evidence from cryptocurrency.</i>	Working paper	<i>SSRN</i>	Mixed
16	He et al. (2022). <i>Fundamentals of perpetual futures.</i>	Working paper	<i>SSRN</i>	Mixed
17	Kim et al. (2023). <i>Pricing power perpetual futures.</i>	Working paper	<i>SSRN</i>	Modeling
18	Phang (2023). <i>Regulating cryptocurrency derivatives: A comparative analysis of approaches in the UK, the EU, and Singapore.</i>	Peer-reviewed	<i>Journal of Business Law</i>	Legal
19	Robertson and Zhang (2025). <i>Price discovery in bitcoin spot and futures markets.</i>	Peer-reviewed	<i>Journal of International Money and Finance</i>	Empirical
20	Ruan and Streltsov (2022). <i>Perpetual futures contracts and cryptocurrency market quality.</i>	Working paper	<i>SSRN</i>	Empirical
21	Shiller (1993). <i>Measuring asset values for cash settlement in derivative markets: Hedonic repeated-measures indices and perpetual futures.</i>	Peer-reviewed	<i>The Journal of Finance</i>	Modeling
22	Werapun et al. (2025). <i>Exploring risk and return profiles of funding rate arbitrage on CEX and DEX.</i>	Peer-reviewed	<i>Blockchain: Research and Applications</i>	Empirical

Conceptual foundations and taxonomy

The concept of a perpetual futures contract predates cryptocurrency markets (Shiller, 1993; Angeris et al., 2023; Neubert et al., 2026). Shiller (1993) proposed perpetual futures as a way to create tradable claims on illiquid or hard-to-price underlying variables. In that original formulation, perpetual contracts addressed a price discovery problem (Shiller, 1993; Angeris et al., 2023; Neubert et al., 2026). Cryptocurrency markets operationalized the idea in a different environment (Shiller, 1993; Angeris et al., 2023; Neubert et al., 2026). Here, the underlying asset is tradable, but traders demand a derivative that offers continuous leveraged exposure without contract rollover (Shiller, 1993; Angeris et al., 2023; Neubert et al., 2026). The resulting contract is best understood as a futures-like instrument without an expiry, anchored to the spot through recurring funding payments rather than through final settlement (Shiller, 1993).

The literature largely agrees on this core logic but diverges on taxonomy (Phang, 2023; Dell’Erba & Vianelli, 2025). Phang (2023) and Dell’Erba and Vianelli (2025) show that a perpetual can be framed as a futures contract, a swap, a rolling contract-for-difference-like instrument, or a hybrid derivative depending on settlement, margin, and regulatory context. This ambiguity is not semantic noise (Phang, 2023; Dell’Erba & Vianelli, 2025). Legal classification affects market access, disclosure obligations, prudential treatment, and cross-border enforcement (Phang, 2023; Dell’Erba & Vianelli, 2025).

Within the finance literature, four contract types are central (Ackerer et al., 2025; Atzberger et al., 2024; Clark, 2023; Kim et al., 2023; Neubert et al., 2026). Linear perpetuals produce a payoff that is linear in the underlying price and are usually margined and settled in a stablecoin or fiat-linked unit (Ackerer et al., 2025; Atzberger et al., 2024; Clark, 2023; Kim et al., 2023; Neubert et al., 2026). Inverse perpetuals quote the contract in fiat terms but settle in the

underlying cryptocurrency, which creates an asymmetric payoff and margin profile (Ackerer et al., 2025; Atzberger et al., 2024; Clark, 2023; Kim et al., 2023; Neubert et al., 2026). Quanto perpetuals separate the pricing unit, the collateral asset, and the settlement asset (Ackerer et al., 2025; Atzberger et al., 2024; Clark, 2023; Kim et al., 2023; Neubert et al., 2026). Power perpetuals extend the same funding-based logic to non-linear payoffs and can be read as an attempt to concentrate option-like exposure in a single perpetual instrument (Ackerer et al., 2025; Clark, 2023; Kim et al., 2023). Related non-linear perpetual structures extend the non-expiring logic further, but the evidence base identified in this review is still much thinner for those structures than for standard linear, inverse, and quanto perpetuals (Clark, 2023; Kim et al., 2023).

The most useful conceptual distinction is therefore not simply between futures and swaps, but between expiring and non-expiring derivative structures (Neubert et al., 2026; Angeris et al., 2023). The defining innovation is the replacement of terminal convergence with periodic funding transfers (Neubert et al., 2026; Angeris et al., 2023). Once that point is clear, the rest of the literature becomes easier to organize. A complementary working paper by Neubert, Rams, and Gruhn (2026) reaches a similar conclusion and treats perpetual futures and swaps as a distinct class of funding-driven, non-expiring crypto-derivatives.

Pricing, funding, no-arbitrage, and basis

Pricing theory is the most mature strand of the literature (Ackerer et al., 2025; Angeris et al., 2023; He et al., 2022). The central question is simple: How can a contract without expiry have a well-defined no-arbitrage value? Recent work provides a substantially more coherent answer (Ackerer et al., 2025; Angeris et al., 2023; He et al., 2022). Angeris et al. (2023) offer a compact primer that clarifies the mechanics of perpetuals and their funding-based anchoring

logic. Ackerer et al. (2025) provide a strong peer-reviewed benchmark for current perpetual-pricing theory. They derive no-arbitrage prices for linear, inverse, and quanto perpetuals in discrete and continuous time and show that the perpetual price can be expressed as the risk-neutral expectation of the spot price sampled at a random time linked to funding intensity (Ackerer et al., 2025; Angeris et al., 2023; He et al., 2022).

This random-time intuition matters because it connects perpetual pricing to standard futures theory while preserving the contract's non-expiring character (Ackerer et al., 2025; He et al., 2022). He et al. (2022) reach a related result through a random-maturity arbitrage framework. Their contribution is especially useful because it also derives no-arbitrage bounds in the presence of trading costs and empirically shows that deviations from benchmark values can be large even when the mean deviation is modest (Ackerer et al., 2025; He et al., 2022).

The literature also converges on the importance of contract form (Ackerer et al., 2025; Atzberger et al., 2024; Cheng et al., 2021). Linear, inverse, and quanto perpetuals are not interchangeable for pricing purposes (Ackerer et al., 2025; Atzberger et al., 2024; Cheng et al., 2021). Inverse contracts embed the margin asset into the payoff structure, thereby changing the relationship among price, collateral, and liquidation risk (Ackerer et al., 2025; Atzberger et al., 2024; Cheng et al., 2021). Quanto contracts add currency or asset dimension, creating convexity effects and complicating hedging (Ackerer et al., 2025; Atzberger et al., 2024; Cheng et al., 2021). Atzberger et al. (2024) and Ackerer et al. (2025) both illustrate why these design choices matter for valuation and strategy.

Where the literature remains unsettled is funding-rule design (Ackerer et al., 2025; He et al., 2022). Ackerer et al. (2025) show that some funding specifications can guarantee exact spot–futures coincidence under idealized conditions. He et al. (2022) show that real exchanges

operate with clamps, caps, and exchange-specific formulas that trade off convergence against manipulation resistance and market stability. A practitioner illustration of BTC perpetual swaps also shows that contemporaneous funding rates can differ materially across major venues and even vary in sign, underscoring the importance of exchange-specific funding rules (Shi et al., 2024). The field, therefore, treats funding as the core anchoring mechanism, but it has not yet identified a single funding rule that is optimal under realistic frictions (Ackerer et al., 2025; He et al., 2022).

Perpetuals were designed to minimize basis risk relative to dated futures, and the literature generally indicates that they do so (De Blasis & Webb, 2022; Christin et al., 2023; Gornall et al., 2025; Werapun et al., 2025). Yet the basis does not disappear (De Blasis & Webb, 2022; Christin et al., 2023; Gornall et al., 2025; Werapun et al., 2025). In perpetual markets, basis is not a simple time-to-maturity object (De Blasis & Webb, 2022; Christin et al., 2023; Gornall et al., 2025; Werapun et al., 2025). It is a moving equilibrium outcome shaped by funding incentives, arbitrage capital, market segmentation, and speculative demand (De Blasis & Webb, 2022; Christin et al., 2023; Gornall et al., 2025; Werapun et al., 2025). This difference explains why the basic literature is central rather than peripheral to understanding perpetuals (De Blasis & Webb, 2022; Christin et al., 2023; Gornall et al., 2025; Werapun et al., 2025).

Peer-reviewed and working-paper evidence reaches a similar qualitative conclusion: contract design shapes arbitrage and spot alignment (De Blasis & Webb, 2022; Christin et al., 2023; Gornall et al., 2025; Werapun et al., 2025). De Blasis and Webb (2022) demonstrate this directly. Christin et al. (2023) document a sizeable crypto carry trade based on short-perpetual and long-spot positions and interpret the resulting returns as compensation for leverage demand and limited arbitrage capital. Gornall et al. (2025) deepen this point by comparing quarterly

futures with perpetuals and arguing that perpetuals dominate because frequent funding payments reduce extreme price dislocations and lower the drawdowns of basis trades. Werapun et al. (2025) extend the discussion of funding-rate arbitrage to both CEX and DEX settings and show that risk and return profiles vary materially with venue design and leverage.

Theoretical and empirical strands line up here (Christin et al., 2023; Gornall et al., 2025; Werapun et al., 2025). Funding creates convergence incentives, but harvesting those incentives is not costless (Christin et al., 2023; Gornall et al., 2025; Werapun et al., 2025). Arbitrageurs must finance spot positions, post margin, manage liquidation risk, and tolerate the possibility that basis dislocations widen before they mean-revert (Christin et al., 2023; Gornall et al., 2025; Werapun et al., 2025). The basis remains economically meaningful even in a contract designed to minimize it (Christin et al., 2023; Gornall et al., 2025; Werapun et al., 2025).

The literature has also widened the meaning of arbitrage beyond spot-versus-perpetual trades (Alexander et al., 2024). Alexander et al. (2024) show that option-perpetual cross-market inefficiencies can also be substantial, especially when inverse contracts and fiat-free parity relations are involved. This result reinforces a broader point: perpetuals are now central to the whole crypto-derivatives ecosystem, not only to directional trading (Alexander et al., 2024).

Market quality, liquidity, microstructure, and price discovery

The market-quality literature is more mixed (De Blasis & Webb, 2022; Gornall et al., 2025; Ruan & Streltsov, 2022). A recurring finding is that perpetuals increase trading activity and improve the practical usability of derivatives markets (De Blasis & Webb, 2022; Gornall et al., 2025; Ruan & Streltsov, 2022). The disagreement concerns whether they improve liquidity in a strong sense or whether they increase adverse selection and informational risk (De Blasis & Webb, 2022; Gornall et al., 2025; Ruan & Streltsov, 2022). That disagreement is substantive

rather than semantic (De Blasis & Webb, 2022; Gornall et al., 2025; Ruan & Streltsov, 2022). It goes to the core question of whether perpetuals make spot markets more stable or more fragile (De Blasis & Webb, 2022; Gornall et al., 2025; Ruan & Streltsov, 2022).

De Blasis and Webb (2022) show that contract design matters for liquidity and arbitrage. Working-paper evidence by Gornall et al. (2025) argues that perpetuals are more liquid than quarterly futures, exhibit tighter spreads, and remain more resilient during market stress. That paper interprets the perpetual design as an institutional response to constrained arbitrage capital and volatile speculative demand (De Blasis & Webb, 2022; Gornall et al., 2025).

A different picture emerges from Ruan and Streltsov (2022). Using high-frequency order-book data and exploiting the eight-hour funding cycle, contract introductions, and a natural experiment from Huobi's termination of perpetual trading, they show that perpetual contracts increase spot trading volume while widening quoted spreads (Ruan & Streltsov, 2022). They interpret this pattern as evidence of stronger informed trading around funding windows, which induces market makers to widen spreads to protect against adverse selection (Ruan & Streltsov, 2022). This result does not imply that perpetuals reduce market quality in every sense (Ruan & Streltsov, 2022). Instead, it suggests that greater activity and greater informational toxicity can coexist (Ruan & Streltsov, 2022).

This is a useful reconciliation of the literature (De Blasis & Webb, 2022; Gornall et al., 2025; Ruan & Streltsov, 2022). Perpetuals may deepen trading and compress basis risk while still widening certain spread-based liquidity measures (De Blasis & Webb, 2022; Gornall et al., 2025; Ruan & Streltsov, 2022). The effect depends on which dimension of market quality is under study and on whether the relevant benchmark is volume, spread, resilience, or information asymmetry (De Blasis & Webb, 2022; Gornall et al., 2025; Ruan & Streltsov, 2022).

The price-discovery literature is smaller than the pricing literature, but already points to an important conclusion (Alexander et al., 2020; Robertson & Zhang, 2025; Ruan & Streltsov, 2022). Perpetuals are not only passive mirrors of the spot market (Alexander et al., 2020; Robertson & Zhang, 2025; Ruan & Streltsov, 2022). In many settings, they also help lead price discovery (Alexander et al., 2020; Robertson & Zhang, 2025; Ruan & Streltsov, 2022). This is economically intuitive. Perpetuals concentrate leverage, are easy to short, and often attract better-informed or faster-moving traders than the underlying spot venues (Alexander et al., 2020; Robertson & Zhang, 2025; Ruan & Streltsov, 2022).

Alexander et al. (2020) show that BitMEX perpetuals lead price discovery relative to major bitcoin spot exchanges. Robertson and Zhang (2025) also report that bitcoin spot and futures markets share price discovery, although their design is broader than perpetuals alone. Together with newer microstructure evidence, these findings suggest that the informational role of perpetuals depends on the market regime, exchange dominance, and trader composition (Alexander et al., 2020; Robertson & Zhang, 2025; Ruan & Streltsov, 2022). The literature therefore supports a conditional rather than universal claim: perpetuals often contribute materially to price discovery, but the strength of that effect varies across assets and venues (Alexander et al., 2020; Robertson & Zhang, 2025; Ruan & Streltsov, 2022).

This theme also connects back to market quality (Alexander et al., 2020; Robertson & Zhang, 2025; Ruan & Streltsov, 2022). If perpetuals lead price discovery because informed traders prefer them, then their benefits for informational efficiency may be accompanied by higher adverse-selection costs in related markets (Alexander et al., 2020; Robertson & Zhang, 2025; Ruan & Streltsov, 2022). That is exactly the tension highlighted by the newer

microstructure evidence (Alexander et al., 2020; Robertson & Zhang, 2025; Ruan & Streltsov, 2022).

Leverage, liquidation, hedging, and risk management

The strongest empirical convergence outside pricing appears in the literature on leverage, liquidation, hedging, and margin (Cheng et al., 2021; Alexander et al., 2023; Atzberger et al., 2024; Deng et al., 2021). Here, the message is clear: liquidation mechanics are not a secondary implementation detail (Cheng et al., 2021; Alexander et al., 2023; Atzberger et al., 2024; Deng et al., 2021). They are a first-order economic feature of perpetual markets (Cheng et al., 2021; Alexander et al., 2023; Atzberger et al., 2024; Deng et al., 2021). Any serious analysis of perpetuals must therefore treat exchange margin rules, liquidation thresholds, and collateral choice as endogenous determinants of market outcomes (Cheng et al., 2021; Alexander et al., 2023; Atzberger et al., 2024; Deng et al., 2021).

Cheng et al. (2021) provide early evidence on this point. Using BitMEX data, they show that forced liquidations are economically large, that traders who are liquidated tend to use very high leverage, and that prudent margin requirements should be much higher than the low exchange minimums often advertised to users (Cheng et al., 2021). Their generalized extreme value approach also serves as a reminder that tail risk, not just average volatility, drives optimal margin design (Cheng et al., 2021).

Alexander et al. (2023) push the analysis further by modeling optimal hedging under automatic liquidation and leverage choice. Their central result is that the best hedge is not the one that minimizes variance alone (Alexander et al., 2023). It must also account for the probability of forced liquidation under insufficient collateral (Alexander et al., 2023). This reframing is important because it shows that perpetual hedging differs fundamentally from

conventional futures hedging (Alexander et al., 2023). A hedger can be directionally correct and still fail if the position is liquidated before convergence occurs (Alexander et al., 2023).

Atzberger et al. (2024) make a more strategy-oriented contribution by showing that hedged quanto perpetual strategies can provide lower-volatility exposure under certain market conditions. Deng et al. (2021) also show, in a broader inverse-futures setting that includes perpetuals, that optimal trading with inverse futures can outperform a long-only spot strategy under appropriate utility criteria. Across these studies, the common conclusion is that perpetuals can be effective risk-management tools, but only when leverage and liquidation are modeled explicitly (Atzberger et al., 2024; Deng et al., 2021).

Trader behavior and exchange design

Research on perpetual-specific trader behavior remains comparatively thin (Chen et al., 2024; Christin et al., 2023; He et al., 2022). Within the included perpetual-specific evidence base, behavior is analyzed mainly through speculative demand, funding, exchange architecture, and open-interest dynamics rather than through a comprehensive trader-level taxonomy (Chen et al., 2024; Christin et al., 2023; He et al., 2022). Perpetual-specific studies therefore need to bridge the remaining gap between trader behavior and the institutional details of leverage, funding, and exchange design (Chen et al., 2024; Christin et al., 2023; He et al., 2022).

Christin et al. (2023) provide an informative window into speculative demand. Their carry-trade evidence implies persistent demand to hold leveraged long positions through perpetuals and suggests that arbitrage capital is insufficient to eliminate the resulting premium (Christin et al., 2023). He et al. (2022) similarly report substantial and persistent deviations from benchmark no-arbitrage prices, with comovement across currencies and high-Sharpe arbitrage opportunities.

Chen et al. (2024) provide a developed analysis of market design in this area. They compare centralized and decentralized perpetual markets and classify exchange architectures into centralized, hybrid, oracle-pricing, and virtual automated market maker models (Chen et al., 2024). Their empirical evidence suggests that trader behavior differs materially across architectures (Chen et al., 2024). In VAMM systems, long and short open interest have asymmetric effects on volatility because price formation depends on virtual inventory (Chen et al., 2024). In oracle-based systems, traders behave more like price takers, and long positions appear to accumulate more strongly after positive news (Chen et al., 2024). These results suggest that perpetual-specific behavior is endogenous to exchange design (Chen et al., 2024).

This body of literature is promising but remains sparse (Chen et al., 2024; Christin et al., 2023; He et al., 2022). The current evidence base has not yet produced a comprehensive trader-level behavioral map for perpetual markets (Chen et al., 2024; Christin et al., 2023; He et al., 2022). That gap matters because perpetual markets are where leverage, funding, and liquidation interact most visibly in crypto (Chen et al., 2024; Christin et al., 2023; He et al., 2022).

Regulation, law, and policy

The legal and policy literature is smaller than the finance literature, but it addresses issues that the finance literature often treats as exogenous (Phang, 2023; Dell’Erba & Vianelli, 2025). Phang (2023) shows that cryptocurrency derivatives are regulated quite differently across jurisdictions. The United Kingdom moved toward the prohibition of retail crypto-derivatives (Phang, 2023; Dell’Erba & Vianelli, 2025). The European Union relied more on product intervention and pre-existing derivatives law (Phang, 2023; Dell’Erba & Vianelli, 2025). Singapore adopted a more conditional market-access model (Phang, 2023; Dell’Erba & Vianelli,

2025). These differences are not cosmetic (Phang, 2023; Dell’Erba & Vianelli, 2025). They shape who can trade perpetual-like instruments, where those products are listed, and how exchanges design them (Phang, 2023; Dell’Erba & Vianelli, 2025).

Dell’Erba and Vianelli (2025) extend this analysis, arguing that crypto-derivatives regulation remains fragmented across jurisdictions, even though the market itself is highly cross-border. They identify four drivers of divergence: settlement methods, the legal treatment of the underlying asset, marketability considerations, and broader policy choices (Dell’Erba & Vianelli, 2025). They also stress that crypto derivatives, including perpetuals, now account for most activity in the crypto economy and therefore deserve closer regulatory scrutiny (Dell’Erba & Vianelli, 2025).

The legal literature converges on three concerns (Phang, 2023; Dell’Erba & Vianelli, 2025). First, classification is unstable (Phang, 2023; Dell’Erba & Vianelli, 2025). The same product may be labeled a future, a swap, a CFD-like product, or a bespoke crypto derivative depending on the jurisdiction (Phang, 2023; Dell’Erba & Vianelli, 2025). Second, retail protection remains a dominant policy concern because perpetuals combine leverage, complexity, and opacity (Phang, 2023; Dell’Erba & Vianelli, 2025). Third, fragmentation creates regulatory arbitrage (Phang, 2023; Dell’Erba & Vianelli, 2025). Traders and exchanges can migrate across borders faster than regulators can harmonize standards (Phang, 2023; Dell’Erba & Vianelli, 2025). These concerns make law and policy central to the study of perpetuals rather than an afterthought (Phang, 2023; Dell’Erba & Vianelli, 2025).

Discussion and research agenda

Taken together, the literature suggests four broad propositions (De Blasis & Webb, 2022; Gornall et al., 2025; Ackerer et al., 2025; Cheng et al., 2021; Chen et al., 2024; Phang, 2023;

Dell’Erba & Vianelli, 2025). First, perpetuals appear to have become dominant in part because they reduce rollover frictions and generally track spot more tightly than dated futures (De Blasis & Webb, 2022; Gornall et al., 2025; Akerer et al., 2025; Cheng et al., 2021; Chen et al., 2024; Phang, 2023; Dell’Erba & Vianelli, 2025). Second, pricing theory now provides a credible benchmark for valuing these contracts under funding-based convergence (De Blasis & Webb, 2022; Gornall et al., 2025; Akerer et al., 2025; Cheng et al., 2021; Chen et al., 2024; Phang, 2023; Dell’Erba & Vianelli, 2025). Third, leverage and liquidation are intrinsic to the economics of perpetuals, not implementation noise (De Blasis & Webb, 2022; Gornall et al., 2025; Akerer et al., 2025; Cheng et al., 2021; Chen et al., 2024; Phang, 2023; Dell’Erba & Vianelli, 2025). Fourth, exchange architecture and legal classification materially shape outcomes in these markets (De Blasis & Webb, 2022; Gornall et al., 2025; Akerer et al., 2025; Cheng et al., 2021; Chen et al., 2024; Phang, 2023; Dell’Erba & Vianelli, 2025). This integrated reading is also consistent with Neubert et al. (2026), who systematize the core mechanics of cryptocurrency perpetual futures and swaps in a complementary working paper.

The literature also contains two important tensions (Gornall et al., 2025; Ruan & Streltsov, 2022; Akerer et al., 2025; He et al., 2022; Christin et al., 2023). The first concerns liquidity enhancement versus spread widening (Gornall et al., 2025; Ruan & Streltsov, 2022; Akerer et al., 2025; He et al., 2022; Christin et al., 2023). Perpetuals clearly increase activity (Gornall et al., 2025; Ruan & Streltsov, 2022; Akerer et al., 2025; He et al., 2022; Christin et al., 2023). The open question is whether that activity should be interpreted as improved liquidity or as increased informed trading with higher adverse-selection costs (Gornall et al., 2025; Ruan & Streltsov, 2022; Akerer et al., 2025; He et al., 2022; Christin et al., 2023). The second tension concerns efficiency (Gornall et al., 2025; Ruan & Streltsov, 2022; Akerer et al., 2025; He et al.,

2022; Christin et al., 2023). Pricing theory implies a benchmark, yet working-paper evidence shows persistent deviations from that benchmark and profitable arbitrage opportunities (Gornall et al., 2025; Ruan & Streltsov, 2022; Ackerer et al., 2025; He et al., 2022; Christin et al., 2023). These tensions are not contradictions in the strict sense (Gornall et al., 2025; Ruan & Streltsov, 2022; Ackerer et al., 2025; He et al., 2022; Christin et al., 2023). They are signs that perpetual markets remain shaped by frictions, limited arbitrage capital, exchange-specific rules, and heterogeneous trader demand (Gornall et al., 2025; Ruan & Streltsov, 2022; Ackerer et al., 2025; He et al., 2022; Christin et al., 2023).

Several research gaps remain (Ackerer et al., 2025; He et al., 2022; Chen et al., 2024; Christin et al., 2023; Giagkiozis & Said, 2024). Funding-rule design is still an open problem (Ackerer et al., 2025; He et al., 2022; Chen et al., 2024; Christin et al., 2023; Giagkiozis & Said, 2024). The literature knows that funding matters, but it does not yet know which rule best balances convergence, manipulation resistance, and crisis stability (Ackerer et al., 2025; He et al., 2022; Chen et al., 2024; Christin et al., 2023; Giagkiozis & Said, 2024). Perpetual-specific investor behavior is underdeveloped (Ackerer et al., 2025; He et al., 2022; Chen et al., 2024; Christin et al., 2023; Giagkiozis & Said, 2024). There is not yet enough trader-level evidence on who systematically pays or receives funding, who supplies arbitrage capital, and how liquidation risk shapes beliefs and behavior (Ackerer et al., 2025; He et al., 2022; Chen et al., 2024; Christin et al., 2023; Giagkiozis & Said, 2024). CEX–DEX comparisons also remain narrow, with few platforms and limited time horizons (Ackerer et al., 2025; He et al., 2022; Chen et al., 2024; Christin et al., 2023; Giagkiozis & Said, 2024). Finally, data quality is itself a research problem (Ackerer et al., 2025; He et al., 2022; Chen et al., 2024; Christin et al., 2023; Giagkiozis & Said, 2024). Giagkiozis and Said (2024) show that exchange-reported open interest can be inconsistent

with traded volume, which means that measurement error may distort inference even in high-frequency studies.

These gaps create a clear forward agenda (Ackerer et al., 2025; He et al., 2022; Chen et al., 2024; Phang, 2023). Practitioner risk frameworks already identify continuous trading, crypto collateral, legal enforceability, and market concentration as modeling frictions, which gives this academic agenda additional relevance (Shi et al., 2024). The next generation of work should connect funding-rule design to market quality and liquidation outcomes; compare centralized and decentralized architectures with common metrics; test whether perpetual-induced informed trading spills over into spot markets in bull and bear regimes; and integrate legal classification into empirical work on venue choice, product design, and regulatory migration.

Table 4

Convergence, disagreement, and open gaps

Topic	What the literature now supports	What remains unresolved
Pricing and no-arbitrage	Many studies indicate that perpetual prices depend on spot, funding design, and financing conditions, and that random-maturity or equivalent pricing logic is analytically useful (Ackerer et al., 2025; Angeris et al., 2023; He et al., 2022).	The optimal funding rule under realistic frictions remains unresolved (Ackerer et al., 2025; He et al., 2022).
Basis and arbitrage	Perpetuals often track the spot price more closely than dated futures, but basis trades remain profitable when arbitrage capital is constrained (Christin et al., 2023; De Blasis & Webb, 2022; Gornall et al., 2025; Werapun et al., 2025).	The durability of these profits as markets mature is still uncertain (Christin et al., 2023; Gornall et al., 2025).
Market quality	Many studies find that perpetuals increase trading activity and can strengthen price	The literature disagrees on whether they improve liquidity

	discovery (Alexander et al., 2020; De Blasis & Webb, 2022; Gornall et al., 2025; Robertson & Zhang, 2025; Ruan & Streltsov, 2022).	in a strong sense or widen spreads through informed trading (Gornall et al., 2025; Ruan & Streltsov, 2022).
Leverage and liquidation	Leverage choice and liquidation rules are first-order determinants of welfare and hedge design (Alexander et al., 2023; Atzberger et al., 2024; Cheng et al., 2021; Deng et al., 2021).	There is no consensus on exchange-optimal margin schedules outside single-venue studies (Alexander et al., 2023; Cheng et al., 2021).
Behavior and design	Exchange architecture shapes trading behavior and the relation between open interest, volatility, and funding (Chen et al., 2024; Christin et al., 2023; He et al., 2022).	Trader-level evidence is still sparse, especially for DEX perpetuals (Chen et al., 2024; Christin et al., 2023).
Regulation	Legal classification is fragmented, and retail protection is a persistent concern (Dell’Erba & Vianelli, 2025; Phang, 2023).	A harmonized cross-border framework does not yet exist (Dell’Erba & Vianelli, 2025; Phang, 2023).

Conclusion

This article reviewed the current state of research on cryptocurrency perpetual futures and perpetual swaps. The evidence indicates that the field has moved beyond descriptive novelty (Ackerer et al., 2025; Cheng et al., 2021; De Blasis & Webb, 2022; Alexander et al., 2020; Phang, 2023; Dell’Erba & Vianelli, 2025). Pricing theory is increasingly coherent (Ackerer et al., 2025; Cheng et al., 2021; De Blasis & Webb, 2022; Alexander et al., 2020; Phang, 2023; Dell’Erba & Vianelli, 2025). Risk-management evidence is substantial (Ackerer et al., 2025; Cheng et al., 2021; De Blasis & Webb, 2022; Alexander et al., 2020; Phang, 2023; Dell’Erba & Vianelli, 2025). Market-microstructure work is becoming sharper and more causal (Ackerer et al., 2025; Cheng et al., 2021; De Blasis & Webb, 2022; Alexander et al., 2020; Phang, 2023;

Dell’Erba & Vianelli, 2025). Legal scholarship identifies fragmentation and regulatory arbitrage as structural features of the market (Ackerer et al., 2025; Cheng et al., 2021; De Blasis & Webb, 2022; Alexander et al., 2020; Phang, 2023; Dell’Erba & Vianelli, 2025).

The central conclusion is that perpetuals addressed a real economic problem in crypto markets (De Blasis & Webb, 2022; Gornall et al., 2025; Cheng et al., 2021; Ruan & Streltsov, 2022; Chen et al., 2024; Phang, 2023). They gave traders continuous leveraged exposure without rollover and generally improved spot tracking relative to dated futures (De Blasis & Webb, 2022; Gornall et al., 2025; Cheng et al., 2021; Ruan & Streltsov, 2022; Chen et al., 2024; Phang, 2023). But they did not eliminate fragility (De Blasis & Webb, 2022; Gornall et al., 2025; Cheng et al., 2021; Ruan & Streltsov, 2022; Chen et al., 2024; Phang, 2023). They relocated fragility into other channels: funding windows, margin choice, liquidation cascades, informed trading, exchange design, and cross-border regulation (De Blasis & Webb, 2022; Gornall et al., 2025; Cheng et al., 2021; Ruan & Streltsov, 2022; Chen et al., 2024; Phang, 2023). Future theory, empirical testing, and policy design should start from that trade-off.

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Declarations

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Data Availability and Supplementary Material

All data generated and analyzed during this study and supplementary material are available upon reasonable request.

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The author confirms that this research has not been published previously and is not under consideration for publication elsewhere.

Ethics Statement

This study was conducted in line with the European Code of Conduct for Research Integrity and the data protection principles set out in Regulation (EU) 2016/679 (All European Academies, 2023; European Union, 2016).

Responsible AI Ethics Statement

This study used artificial intelligence (AI) tools to support tasks such as identifying relevant literature, analyzing datasets, translating, and editing textual content. These tools were employed solely to enhance efficiency, and their outputs were critically reviewed to ensure alignment with research objectives. The EU AI Act, the OECD Recommendation on Artificial Intelligence, and the UNESCO Recommendation on the Ethics of Artificial Intelligence (European Union, 2024; Organization for Economic Co-operation and Development, 2019; United Nations Educational, Scientific and Cultural Organization, 2021) guided the use of AI. The authors made all final decisions and retain full responsibility for the integrity, rigor, and conclusions of this research.

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